Q-4) Discuss SAR of alcohols & their derivatives

ALCOHOLS

sedative hypnotics (miscellaneous)

Actions: Acts at the GABAA receptor to potentiate the action of GABA.

Also inhibits NMDA (N-Methyl-D-Asparate) glutamate receptors.

These actions result in an array of actions including CNS depression.

SAR for hypnotic activity of alcohols

- Activity increases as the chain length increase with maximum at C8
- Branching the chain increases activity

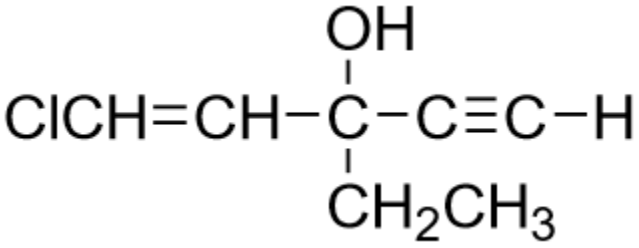
Tertiary > Secondary > Primary

- **decrease metabolism by oxidation to the corresponding carboxylic acids**
- Chlorination or bromination of simple and branched alcohol induces favorabl effect on distribution coefficient. For example trichloro or tribromoethanol has strong hypnotic activity

- **Replacing hydrogen with a halogen increases activity**

Ethchlorvynol and Chlorobutanol

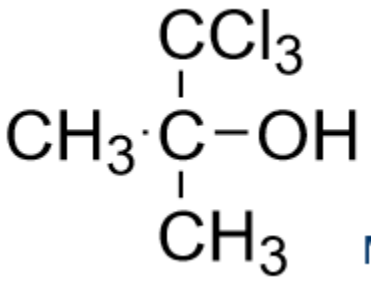
– A tertiary, lipophilic alcohol



Ethchlorvynol (1-chloro-3-ethyl-1-penten-4-yn-3-ol)

Chlorobutanol

- 2 hydroxy, 2 methyl, 1,1,1 trichloropropane (**trichloro-2-methyl-2-propanol)**
- It is a strong hypnotic agent and has been used as preanesthetic medication
- It is as dangerous as chloral hydrate



Chlorobutanol

Methyl parafynol

Has pronounced hypnotic effect with wide margin of safety

Active orally and parenterally

Short duration of action due to oxidation of triple bond



Alcohol derivatives

- ✓ **Carbamates**

Carbamylation increases depressant potency

- **Carbamates-** much more resistant to metabolic inactivation than hydroxyl functions

The best example is Ethinamate which is Carbamate analogue of acetylenic alcohol.

Ethinamate

is a short-acting carbamate-derivative sedative-hypnotic medication used to treat insomnia. Regular use leads to drug tolerance, and it is usually not effective for more than 7 days. Prolonged use can lead to dependency.

Synthesized by combining acetylene with cyclohexanone and the transformation of the resulting carbinol into carbamate by the subsequent reaction with phosgene, and later with ammonia.

